Submission details

Track	4 — Best Agentic Economy Experience on Arc
Participant	Ronie Joseph (solo)
Circle Developer Account	roniejosephv@gmail.com
Circle products used	USDC · Wallets (developer-controlled) · Gateway · Nanopayments (x402)
Live demo (dashboard)	https://autopilotdashboard-production.up.railway.app
SpendGuard API (public)	https://autopilotsigner-production.up.railway.app/summary
GitHub repository	https://github.com/roniejosephv-star/subscription-autopilot
Demo video	*TODO — paste hosted demo-video URL, then regenerate*
On-chain audit anchor	SpendAnchor 0xfe18…e435 on Arc Testnet (arcscan)
Scope	Testnet demo — Arc Testnet, chain 5042002 (gas = USDC)

Working MVP — deployed and verifiable now

Frontend + backend live: the owner dashboard at https://autopilotdashboard-production.up.railway.app renders the production ledger (100+ settled nanopayments), budget burn-down, approval queue, policy chain, and the Gateway treasury balance; the public signer API serves the same data raw. Four services on Railway (SpendGuard signer, 3 x402 sellers, agent, dashboard), SQLite ledger on a persistent volume. Custody mode is SIGNER_MODE=circle — EIP-712 via Circle Wallets DCW EOA, verified against Gateway settlement on Arc Testnet (transfer 9bd6a149-1821-4341-a4aa-712cdc362382, 2026-07-06). Spend epochs auto-anchor on Arc every 10 minutes.

Demo guide (the video's four beats, reproducible live)

1 · Delegation	Agent researches quotes, subscribes to the cheapest seller, meters per-call gasless nanopayments.
2 · Negotiation	Seller price drift → at renewal the agent re-quotes all sellers at real volume and switches (21.9% cheaper per call).
3 · Attack	Prompt-injected overspend → denied at the per-tx policy wall; a $0.03 premium call is held → human Approve/Deny on the dashboard.
4 · Proof	Burn-down, savings, denial log, Gateway receipts — and the epoch chip linking to the SpendAnchor tx on arcscan.

Documentation

README.md — zero-to-running setup and per-product integration notes with file references · ARCHITECTURE.md — component spec and trust boundaries · PRD.md — product spec, acceptance criteria, risk log · DEPLOYMENT.md — the exact deployment runbook used · docs/demo-script.md — video script · docs/submission/ — this document's sources.

Judge quick-verify (no setup): open the dashboard; curl https://autopilotsigner-production.up.railway.app/summary; click the SpendAnchor link on arcscan.

Circle Product Feedback

Why we chose these products

Nanopayments/x402 is the only rail where per-call metered agent payments are economically sane (gasless, sub-cent, batched). Gateway gives the agent a fundable balance with instant receipts. Circle Wallets (DCW EOA) moves key custody out of the agent entirely — the property our whole trust model rests on. USDC-as-gas on Arc collapses faucet/gas UX to one token.

What worked well

• DCW EOA signTypedData signatures are fully ecrecover-compatible with Gateway's x402 settle — the custody-separation design works E2E on Arc Testnet (verified 2026-07-06, transfer 9bd6a149-…-712cdc362382).
• Seller monetization is genuinely 2 lines (createGatewayMiddleware + gateway.require()); per-request dynamic pricing works by binding require(price) at call time — it became our negotiation surface.
• CHAIN_CONFIGS exports eliminated all address/RPC hardcoding.
• Arc deposit finality (~0.5 s) makes the deposit→pay loop feel instant; Wallets contract executions confirm in well under a minute.
• The x402 lifecycle hooks are an excellent extension surface — our entire policy layer lives in one hook.

What could be improved (all found and reproduced during this build)

1. **Nanopayments is EOA-only** (ecrecover on EIP-3009; EIP-1271/SCA unsupported) — easy to miss: Circle's own arc-escrow sample creates accountType SCA wallets, which would silently fail here. Needs a prominent docs callout + runtime hint.
2. **Signature-validity contradiction:** seller quickstart says ≥ 7 days, the SDK error reference says ≥ 3, and the exported GATEWAY_AUTH_VALIDITY_WINDOW_SECONDS constant is undocumented. Pick one number, document the constant.
3. **18-decimal gas vs 6-decimal ERC-20 USDC on Arc** deserves an explicit docs callout — it bites any native-balance math.
4. **Sample apps are heavy references:** arc-escrow needs Supabase + Docker + ngrok + OpenAI before first run; a minimal single-process reference per product would cut onboarding dramatically.
5. **Undocumented header in shipped types:** X-ARC-PRIVATE-MAINNET-ENABLED appears in the SDK's .d.ts, nowhere in docs.
6. **Hook documentation lives off-site** (x402.org, not developers.circle.com) — mirror or link prominently; it's the SDK's best surface.
7. **Broken script in arc-fintech sample:** package.json declares spend:gateway:arc → a file that doesn't exist in the repo.
8. **Wallets ↔ x402 SDK interop wart:** signTypedData requires an explicit EIP712Domain entry in types, but viem-style callers (including BatchEvmScheme's signer interface in Circle's own SDK) omit it. The error — "there is extra data provided in the message (0 < 4)" — names neither the missing type nor the fix. Accept viem-normalized typed data, or return "types.EIP712Domain is required".
9. **GatewayClient.deposit() requires a raw private key,** so DCW-custodied wallets can't use it. Verified workaround: replicate its two calls (USDC.approve + GatewayWallet.deposit) via createContractExecutionTransaction. A documented DCW-native deposit path is the single change that would most help agentic use cases — Circle-custodied buyers are exactly who agents need.

Recommendations

Ship a first-class "agent buyer" recipe: DCW EOA wallet + Gateway deposit via contract execution + x402 client with a policy hook — the five pieces exist today but must be discovered across four doc sites and two SDKs' type definitions. This project is, in effect, that recipe; we'd be glad if it were useful as a reference.